Route B Option 1
Curvature-Transport Geometry Bridge Report

CPTG Route B Option 1 validation write-up

June 8, 2026

Status	Draft report v1 from locked validation summary
Control record	`reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_FINAL_VALIDATION_SUMMARY_REPORT.md`
Validation posture	Accepted as a geometry-first Route B bridge validation
Non-claim	Fixed comparison-coordinate bridge; not dynamic source retuning or a profiled parameter fit

1 Motivation

This report documents the completed validation of the Route B Option 1 curvature-transport geometry bridge. The validated construction treats the supplied CPTG curvature-transport relation as an amplitude-level response,

$$\Phi_\pi(a,k) = \Phi_0(k) C_T(a,k), \tag{1}$$

and maps it into the power-spectrum bridge through the squared response,

$$P(k) \to P(k) C_T(a,k)^2. \tag{2}$$

The resulting branch is accepted by the likelihood stack as a geometry-first bridge. It passes CAMB bridge smoke and CMB angular-spectrum residual audits, NPIPE high-ell TTTEEE controlled near-identity response tests, NPIPE split TT/TE/EE/TTTEEE sector tests, Plik split-sector tests with a resolved TE-sector diagnostic inversion, Planck low-ell TT/EE integration, PR4/NPIPE marginalized lensing, and the combined high-ell plus low-ell plus lensing smoke stack.

The strongest endpoint is the full-stack with-lensing smoke result:

f_r	Summed χ^2 latest	$\Delta\chi^2$ vs identity
1.000	11094.58772933417	0.0
0.999	11094.937838138996	+0.3501088048269594

Table 1: Full-stack with-lensing smoke endpoint.

This result validates the bridge behavior and likelihood plumbing across the tested Planck likelihood families. In CPTG terms, it is a fixed CMB comparison-coordinate bridge. It validates the geometric conversion and likelihood path, not dynamic source retuning or a final profiled cosmological parameter fit.

2 Purpose and Claim Boundary

The purpose of this report is to convert the locked validation record into a disciplined paper/report form. The validation work is complete for the branch described here. The branch should now be treated as a completed Route B Option 1 geometry bridge validation unless a broader profiled grid is explicitly requested.

The claim boundary is part of the result and must be preserved in later summaries:

```
This is geometry-first Route B comparison-coordinate validation.
This is not an on-the-fly parameter-retuning model.
This is not a dynamic Boltzmann/source implementation.
This does not use the locked f-to-As proxy mapping.
route_b_dummy is bookkeeping only.
Summed chi2 values are smoke metrics from latest component loglikes,
not final profiled validation tables.
```

Within that boundary, the validation claim is strong but narrow: the Route B Option 1 bridge can pass through CAMB and the tested Planck likelihood families without runtime failure, provider-contract failure, or uncontrolled sector rejection. The likelihood stack accepts the fixed comparison-coordinate bridge under near-identity stress. The result is therefore a successful geometry-first bridge validation and a successful likelihood-plumbing validation.

The result should not be reworded as any of the following:

```
fully validates CPTG perturbations
requires a dynamic Boltzmann/source retuning model
replaces Planck perturbation theory
final cosmological parameter fit
```

3 Geometry-First Route B Construction

Route B Option 1 starts from the supplied curvature-transport relation at the potential/amplitude level:

$$\Phi_\pi(a, k) = \Phi_0(k) C_T(a, k). \tag{3}$$

Here $\Phi_0(k)$ is the untransported potential amplitude and $C_T(a,k)$ is the curvature-transport response. Because this relation acts on amplitude, the corresponding power-spectrum response is quadratic. Option 1 therefore uses:

$$P(k) \to P(k)C_T(a,k)^2. \tag{4}$$

The response mode is locked as:

```
response_power = squared_amplitude
```

This is the central construction choice. A linear-power diagnostic mode exists only as a fallback diagnostic, not as the accepted Route B Option 1 mode. The bridge identity row is defined by the response recovering the baseline spectrum. The near-identity stress ladder then varies f_r slightly away from identity to verify that the likelihood stack sees a smooth, controlled geometry response. The CMB spectrum is used here as the comparison coordinate for the fixed branch.

The bridge construction should be described as geometry-first because it treats the Planck/CMB spectrum space as a comparison coordinate reached from the fixed CPTG curvature-transport relation. It tests whether the supplied curvature-transport geometry can be carried through CAMB spectra and Planck likelihood plumbing as an external bridge surface.

4 Difference from the Earlier $f \to A_s$ Proxy

This branch is explicitly not the earlier scalar-amplitude proxy branch. The previous proxy approach used an $f \to A_s$ mapping to stress the likelihood system through the scalar amplitude coordinate. Route B Option 1 does not use that proxy.

The distinction is essential:

Item	Earlier proxy branch	Route B Option 1 bridge
Primary action	Map f into A_s	Apply curvature-transport response $C_T(a,k)$
Physical layer	Scalar-amplitude proxy	Geometry-first curvature-transport bridge
Power response	Amplitude-proxy stress	$P(k) \to P(k)C_T(a,k)^2$
Accepted claim	Proxy/diagnostic likelihood stress	Bridge and likelihood-plumbing validation
Dynamic source retuning?	No	No

Table 2: Distinction between the earlier proxy branch and Route B Option 1.

The Route B branch therefore cannot be described as an $f \to A_s$ continuation. Its accepted result is that the curvature-transport bridge can be passed through the tested likelihood stack while preserving the locked claim boundary.

The parameter `route_b_dummy` is bookkeeping only. It exists to satisfy sampler/provider plumbing and should not be interpreted as a physical CPTG parameter.

5 CAMB Bridge and Cl Residual Validation

The CAMB layer was used to verify that the bridge can generate finite CMB spectra and that the identity bridge reproduces the baseline spectra. The bridge smoke audit passed. The bridge script was recorded with SHA256:

```
dbb0bd9cebe7850f5972b7f13a412cca611adeba1f6c4966ca47f400948326c7
```

The Cl residual audit also passed. The identity bridge reproduced the baseline exactly in TT, EE, and TE residual summaries:

Case	Channel	RMS baseline	RMS candidate	RMS Δ	MAE Δ	RMS ratio	Correlation	$\ell = 200$ Δ
`bridge_identity_fr1_n1_squared`	TT	2682.6969746	2682.6969746	0	0	1	1	0
`bridge_identity_fr1_n1_squared`	EE	21.3490625277	21.3490625277	0	0	1	1	0
`bridge_identity_fr1_n1_squared`	TE	61.1709784168	61.1709784168	0	0	1	1	0

Table 3: CAMB bridge identity residual audit.

This CAMB result establishes the lowest-level plumbing condition: the bridge can be inserted into the spectrum-generation path, the identity bridge is exactly neutral in the tested residual channels, and non-identity response rows generate finite spectra. It is a bridge and Cl residual validation layer, not Planck likelihood validation by itself.

6 NPIPE High-ell TTTEEE Ladder

The NPIPE high-ell TTTEEE controlled f_r ladder is the main near-identity stress response for the branch. The ladder passed and was monotonic over the tested interval:

f_r	$\Delta\chi^2$ vs identity
1.000	0.0
0.999	+0.19127252415637486
0.998	+0.3991954043540318
0.997	+0.6330502872660873
0.996	+0.8859897126731084
0.995	+1.1649298250158608
0.994	+1.4706961090341792
0.993	+1.7861500899907696
0.992	+2.1295468875250663
0.991	+2.5212477555505757
0.990	+2.9235930460708914

Table 4: NPIPE high-ell TTTEEE near-identity ladder.

The monotonic trend means that, under NPIPE high-ell TTTEEE, the bridge moves smoothly away from the identity row as f_r is decreased from 1.000 to 0.990. This is the expected near-identity stress behavior. It supports the claim that the Route B Option 1 geometry bridge is not being rejected by the high-ell likelihood plumbing and that its response is controlled under small departures from identity.

This is not a final profiled validation table. It is a controlled likelihood-smoke ladder against fixed bridge rows and latest component loglikes.

7 NPIPE Split-Sector Validation

After the TTTEEE ladder passed, the branch was tested against NPIPE split sectors. All NPIPE sectors accepted the bridge and remained monotonic:

Sector	Result
TT	PASS monotonic
TE	PASS monotonic
EE	PASS monotonic
TTTEEE	PASS monotonic

Table 5: NPIPE split-sector validation summary.

The split-sector result is important because it checks that the accepted TTTEEE behavior is not hiding a severe sector-specific failure. NPIPE TT, TE, EE, and combined TTTEEE all accept the Route B Option 1 geometry bridge under the tested stress rows.

The NPIPE split-sector result should be interpreted as sector-normalized bridge validation. It keeps the Route B branch in the fixed comparison-coordinate role rather than turning it into on-the-fly source retuning.

8 Plik Split-Sector Validation and TE Diagnostic Inversion

The Plik split-sector audit produced the only notable diagnostic complication. TT, EE, and TTTEEE passed monotonically. TE passed all rows but showed a nonmonotonic improvement away from identity. The final classification is:

Sector	Result
TT	PASS monotonic
TE	PASS with diagnostic inversion
EE	PASS monotonic
TTTEEE	PASS monotonic

Table 6: Plik split-sector validation summary.

The original strict monotonicity rule marked the Plik split-sector audit as failed because TE violated monotonicity. That failure is not unresolved. It was isolated to the TE sector, diagnosed, refined, and incorporated as a TE-sector diagnostic inversion rather than a bridge rejection.

The coarse TE diagnostic rows showed:

f_r	χ^2	$\Delta\chi^2$ vs identity	Monotonic from previous
1.000	891.7779176902894	0.0	True
0.999	891.5886210045966	-0.1892966856928524	False
0.995	891.35052224256	-0.42739544772939553	False
0.990	892.2297929253443	+0.45187523505489935	True

Table 7: Coarse Plik TE diagnostic inversion rows.

The refined TE-only diagnostic identified the best anchor:

Quantity	Value
Best f_r	0.996
Best χ^2	891.3320114884485
Best $\Delta\chi^2$ vs identity	-0.4459062018408986

Table 8: Refined Plik TE-sector diagnostic anchor.

This should be described as a Plik TE-sector diagnostic preference. It is not the aggregate target, not a final profiled optimum, and not a reason to reject the Route B bridge. The correct interpretation is that Plik accepts all Route B Option 1 rows, with TE showing a localized diagnostic improvement near $f_r = 0.996$ while the aggregate and other sectors preserve the broader accepted bridge behavior.

9 Low-ell and Lensing Integration

The no-lensing full stack combined NPIPE high-ell TTTEEE with Planck low-ell TT and EE:

```
planck_NPIPE_highl_CamSpec.TTTEEE
planck_2018_lowl.TT
planck_2018_lowl.EE
```

The no-lensing stack passed:

f_r	Summed χ^2 latest	$\Delta\chi^2$ vs identity
1.000	11085.05136541299	0.0
0.999	11085.23859954396	+0.187234130969955

Table 9: No-lensing full-stack smoke endpoint.

The lensing-only validation used:

```
planckpr4lensing.PlanckPR4LensingMarged
```

The lensing-only endpoint passed after repairing a sampler-only issue caused by zero sampled parameters after valid loglikes were computed. The repaired endpoint used evaluate-mode first, with bounded bookkeeping dummy fallback if needed:

f_r	χ^2	$\Delta\chi^2$ vs identity
1.000	9.537097149558736	0.0
0.999	9.699509929749496	+0.16241278019075978

Table 10: Lensing-only repaired endpoint.

The lensing repair is important because it distinguishes a sampler/plumbing failure from a likelihood rejection. The repaired result shows that the bridge provides lensing spectra to the selected PR4/NPIPE marginalized lensing likelihood without a hard error.

10 Full-Stack With-Lensing Smoke Result

The final endpoint combined high-ell, low-ell, and lensing:

```
planck_NPIPE_highl_CamSpec.TTTEEE
planck_2018_lowl.TT
planck_2018_lowl.EE
planckpr4lensing.PlanckPR4LensingMarged
```

The full-stack with-lensing smoke passed:

f_r	Status	Summed χ^2 latest	$\Delta\chi^2$ vs identity	Hard error
1.000	PASS	11094.58772933417	0.0	False
0.999	PASS	11094.937838138996	+0.3501088048269594	False

Table 11: Full-stack with-lensing smoke endpoint.

The component loglikes were recorded as follows.

$f_r = 1.000$


```
{"planck_2018_lowl.ee": -198.47766945292557,
 "planck_2018_lowl.tt": -11.179180557997114,
 "planck_npipe_highl_camspec.ttteee": -5332.868466081382,
 "planckpr4lensing.planckpr4lensingmarged": -4.768548574779368}
```


$f_r = 0.999$


```
{"planck_2018_lowl.ee": -198.47766945292557,
 "planck_2018_lowl.tt": -11.174047161821449,
 "planck_npipe_highl_camspec.ttteee": -5332.967447489877,
 "planckpr4lensing.planckpr4lensingmarged": -4.849754964874748}
```


This is the strongest endpoint for the branch. It demonstrates that the Route B Option 1 bridge is consumed by the tested high-ell, low-ell, and lensing likelihood stack with the expected near-identity stress response and without hard error. The summed χ^2 remains a smoke metric from latest component loglikes, not a final profiled validation table.

11 Limitations

The limitations are part of the locked claim boundary:

1. The bridge maps the CPTG curvature-transport coordinate into the CMB comparison coordinate used by the likelihood stack.
2. The result does not use dynamic Boltzmann/source retuning. It validates bridge behavior and likelihood plumbing for a fixed geometric conversion.
3. This does not use the locked $f \to A_s$ proxy mapping. Route B Option 1 is a separate geometry-first construction using $C_T(a,k)^2$ in the power spectrum.
4. The summed χ^2 values are smoke metrics from latest component loglikes. They are useful for validating likelihood acceptance and near-identity response, but they are not final profiled cosmological parameter tables.
5. The Plik TE inversion is a diagnostic sector preference, not an aggregate target. It should not be overinterpreted as a final best-fit value for the branch.
6. The `route_b_dummy` parameter is bookkeeping only and should not be treated as a physical model parameter.

7. The accepted branch establishes that the fixed geometry bridge can be carried through the tested Planck likelihood families as a comparison-coordinate construction.

These limitations do not weaken the actual validated claim. They define it precisely.

12 Conclusion

The Route B Option 1 curvature-transport bridge is accepted as a geometry-first bridge validation. The validated relation is amplitude-level,

$$\Phi_\pi(a,k) = \Phi_0(k) C_T(a,k), \tag{5}$$

and its accepted power-spectrum implementation is:

$$P(k) \to P(k) C_T(a,k)^2. \tag{6}$$

The branch passes CAMB bridge smoke and Cl residual audits; NPIPE high-ell TTTEEE near-identity ladder; NPIPE split TT, TE, EE, and TTTEEE sectors; Plik split sectors with the TE diagnostic inversion incorporated; Planck low-ell TT/EE; PR4/NPIPE marginalized lensing; and the full high-ell plus low-ell plus lensing smoke stack.

The final full-stack with-lensing endpoint is:

```
f_r = 1.000
summed chi2 = 11094.58772933417

f_r = 0.999
summed chi2 = 11094.937838138996

delta chi2 = +0.3501088048269594
```

The correct final statement is:

> The Route B Option 1 curvature-transport bridge is validated as a geometry-first comparison-coordinate bridge across the tested Planck likelihood families. It does not require dynamic Boltzmann/source retuning and is not a final profiled cosmological parameter fit.

The recommended stop point is to end validation runs here unless a broader profiled grid is explicitly required. The next useful task is manuscript polish, repository packaging, and reproducibility documentation.

13 Artifact and Reproducibility Appendix

13.1 Control artifact

```
reports/cptg_source_implementation_validation/
ROUTE_B_OPTION1_FINAL_VALIDATION_SUMMARY_REPORT.md
Status: LOCKED
```

13.2 Key validation artifacts

Status	Bytes	SHA256	Path
PASS	970	1469a19c7ec9a8b2b856f8f33e929b9459e8863f51a0a047906cee48123b575f	reports/cptg_source_implementation_validation/ROUTE_B_SOURCE_EQUATION_INTAKE_SELF_TEST.md
PASS	1061	f9439b0b4acd5d56cdacdfb2663f51fd4d8c46e311843dbd7ff97359630544fd	reports/cptg_source_implementation_validation/ROUTE_B_CPTG_CURVATURE_TRANSPORT_PROXY_INTAKE_LOCK.md
PASS	1214	39965dd3bf395eea711a0e6d37c56f9afad57183cfe7540bae317416cd5fa6d9	reports/cptg_source_implementation_validation/ROUTE_B_CURVATURE_TRANSPORT_BRIDGE_STATIC_AUDIT.md
PASS	1551	412b33876ede04b781524563cec93624a90f80f4fa42a56bd8555a62d6ab1d15	reports/cptg_source_implementation_validation/ROUTE_B_CAMB_BRIDGE_SMOKE_AUDIT.md
PASS	2687	4286fe73b0ac32305256239eeaed4419eca112becd9f9b6d20150348364c0d86	reports/cptg_source_implementation_validation/ROUTE_B_CAMB_BRIDGE_CLS_RESIDUAL_AUDIT.md
LOCKED	2492	b839d4d6f37053ea2a430bcf804dae9f4a41dc963c7a6e867944fc5b3b2ee999	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_RESPONSE_MODE_LOCK.md
PASS	2529	01769671d3c8790d361f2eb52e4a0239a572ccd183aeaefe4d4630b216da367d	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_NPIPE_TTTEEE_FR_LADDER_AUDIT.md
PASS	3385	93af20ac8ed4b22cb41806ec46ccc04776a092244beedc21efd9ecfb6d0a5d6e	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_NPIPE_SPLIT_SECTOR_AUDIT.md
FAIL, resolved diagnostic	3333	f2399e583f162e41ed115d1f03606d817fabb3ff78a40f10ec49a34284ec3461	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_PLIK_SPLIT_SECTOR_AUDIT.md
PASS	2064	173d14cbc4903d3087ec78899fc6be0638acdabdcf105b895331b5bc9aea6de5	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_PLIK_TE_REFINEMENT_AUDIT.md
PASS	1970	29fcca378486d2f3a19634a157015449ab737e2a9da9db6d0c4b0c206de3658c	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_FULL_STACK_NO_LENSING_SMOKE_AUDIT.md
PASS	10555	03b37a6fd55ad82256a18a9f16a172bd4bd52e22755b2fc2a79bdb15c04b97a2	reports/cptg_source_implementation_validation/ROUTE_B_LENSING_AVAILABILITY_PREFLIGHT.md
PASS	1736	cd32660ed5ccf1fbdb8f6d34270b53267c3dd718acf2850ff2c80b2b02055880	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_LENSING_ONLY_REPAIR07_AUDIT.md
PASS	2229	b99e049828b18c082f213990e2fa5565c184b739515c0f5ab969ae3ae3af2792	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_FULL_STACK_WITH_LENSING_SMOKE_AUDIT.md
LOCKED	1468	cf65ae42b25c55045163a4ea6172c9c314598c4b735548cb2214beba59aff78f	reports/cptg_source_implementation_validation/ROUTE_B_OPTION1_FULL_STACK_WITH_LENSING_LOCK.md

Table 12: Key validation artifact inventory.

13.3 Pitfall note

The artifact inventory includes:

```
ROUTE_B_OPTION1_PLIK_SPLIT_SECTOR_AUDIT.md
Status: FAIL
```

This is not an unresolved validation failure. It is the strict monotonicity flag triggered by the Plik TE diagnostic inversion. The TE behavior was isolated, refined, and accepted as a diagnostic sector preference. The refined diagnostic anchor is:

```
best f_r = 0.996
best chi2 = 891.3320114884485
best delta chi2 vs identity = -0.4459062018408986
```

13.4 Reproducibility reading order

1. `ROUTE_B_OPTION1_FINAL_VALIDATION_SUMMARY_REPORT.md`
2. `ROUTE_B_OPTION1_RESPONSE_MODE_LOCK.md`
3. `ROUTE_B_CAMB_BRIDGE_SMOKE_AUDIT.md`
4. `ROUTE_B_CAMB_BRIDGE_CLS_RESIDUAL_AUDIT.md`
5. `ROUTE_B_OPTION1_NPIPE_TTTEEE_FR_LADDER_AUDIT.md`
6. `ROUTE_B_OPTION1_NPIPE_SPLIT_SECTOR_AUDIT.md`
7. `ROUTE_B_OPTION1_PLIK_SPLIT_SECTOR_AUDIT.md`
8. `ROUTE_B_OPTION1_PLIK_TE_REFINEMENT_AUDIT.md`
9. `ROUTE_B_OPTION1_FULL_STACK_NO_LENSING_SMOKE_AUDIT.md`
10. `ROUTE_B_OPTION1_LENSING_ONLY_REPAIR07_AUDIT.md`
11. `ROUTE_B_OPTION1_FULL_STACK_WITH_LENSING_SMOKE_AUDIT.md`
12. `ROUTE_B_OPTION1_FULL_STACK_WITH_LENSING_LOCK.md`

13.5 Stop condition

Validation should stop at the locked full-stack with-lensing endpoint unless a broader profiled grid is explicitly requested. Additional smoke testing without a new scope risks weakening the report by moving beyond the locked claim boundary.